Basic Principles of Medicine 2

DM: Digestive System & Metabolism | Lecture 11

Clinical Gastrointestinal System Flipped Classroom

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Achalasia

Failure of Relaxation of the **Cardioesophageal Sphincter**

- Loss of **peristalsis**
- Loss of **myenteric** plexus
- **Dysphagia**
- **Regurgitation**
- **Retrosternal pain**
- **Weight loss**

Chest X-ray shows **dilated esophagus**, **double** contoured **mediastinal widening**.

Case 1

A 44-year-old man comes to the emergency department because of acute onset abdominal pain, nausea, vomiting and generalized weakness. Pain is localized to the epigastric region with radiation to the back and is described as a burning sensation. Patient has a history of smoking and heavy alcohol use. Blood pressure is 112/90mmHg, pulse is 120/min, respirations are 26/min and temperature is 38.4^{o}C (101^{o}F). Physical exam shows a distressed patient with a rigid abdomen and epigastric tenderness.

Labs show increased white blood cells. X-ray is shown. Following stabilization, CT confirms radiographic findings and shows free fluid all over the abdominal cavity. During exploratory laparotomy, purulent fluid with undigested products was seen in all four quadrants. After removal of fluid, a perforation in the first part of the duodenum, just distal to pylorus was seen. Surgery was completed with no post-op complications

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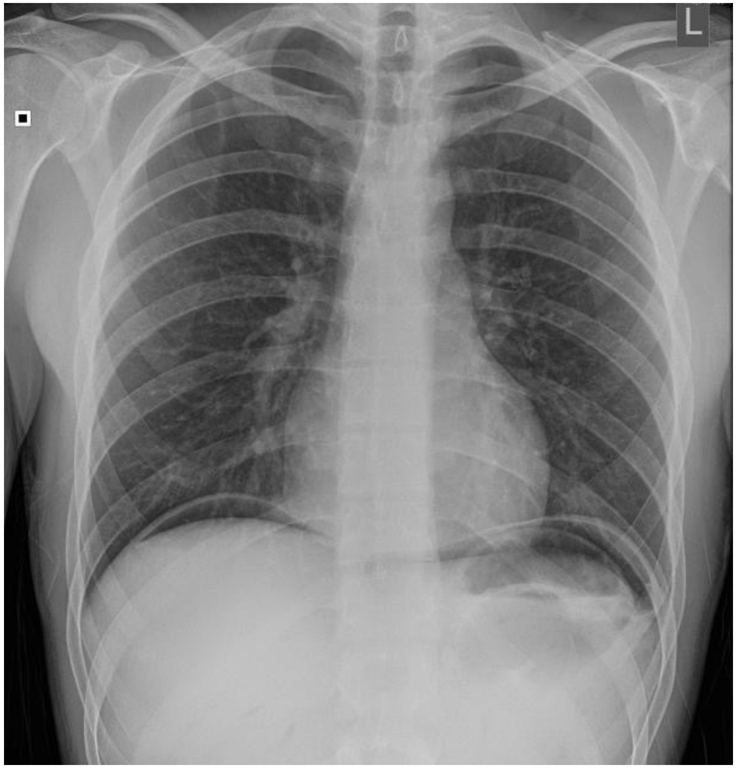


Case Questions

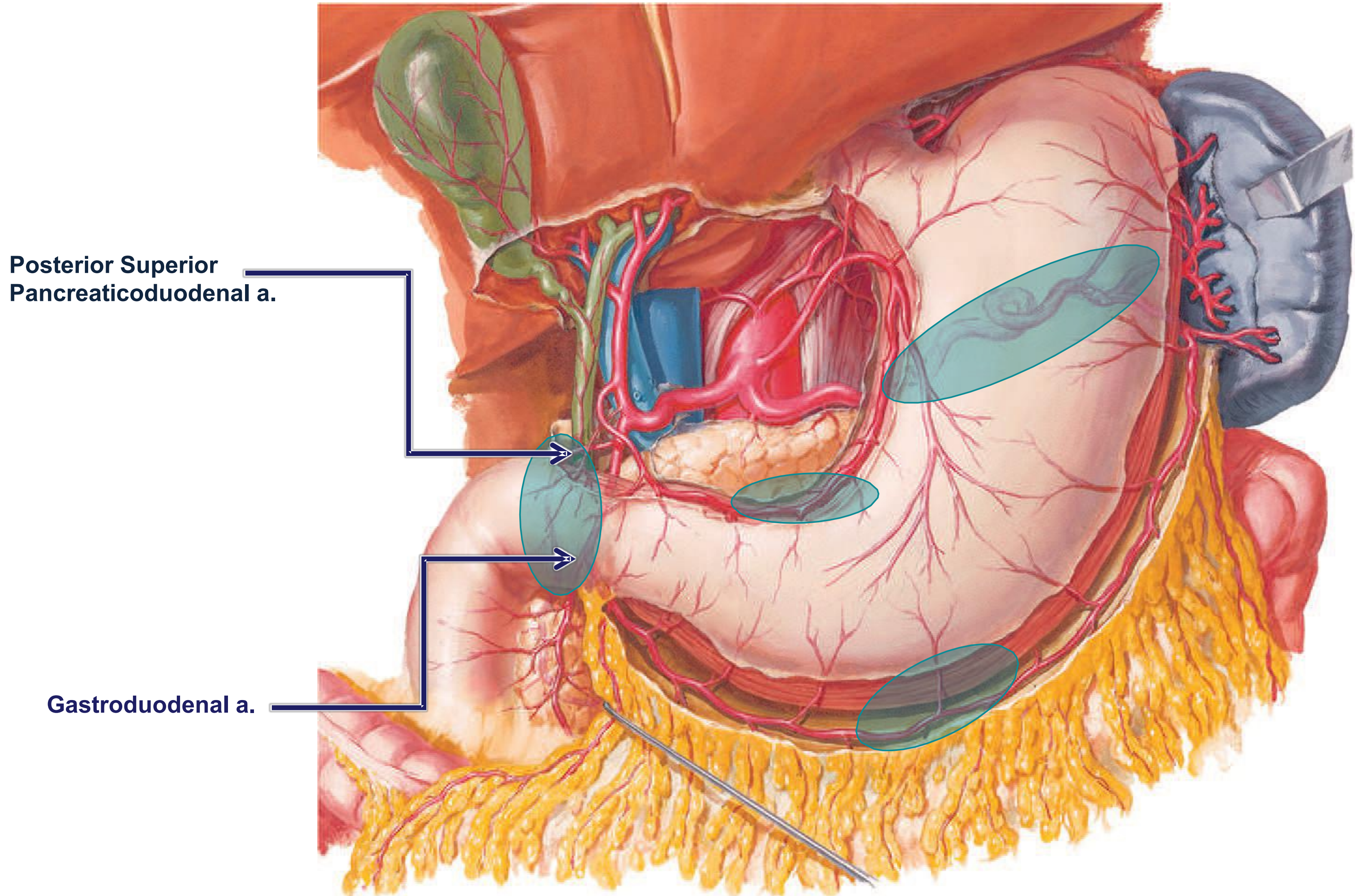
1. Which blood vessel is most at risk for injury by a perforated ulcer in the posterior wall of the duodenal bulb?

2. Which blood vessel(s) will most likely be eroded on ulceration of the posterior wall of the stomach?

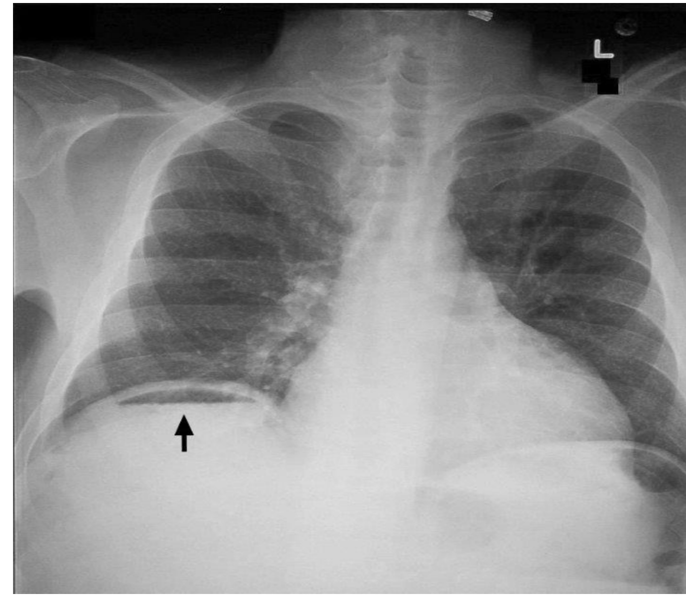
3. Where would the stomach contents from a perforation of the following areas most likely accumulate first?
   a) Posterior stomach wall
   b) Anterior stomach wall
   c) Which most likely occurred in the patient in Case 1 (assuming no changes in position)?

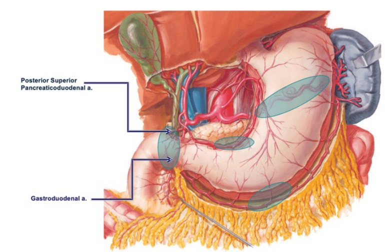
4. Where will free air most likely accumulate in the peritoneal cavity?

5. Describe the pathway of fluid around the abdominal cavity.


Posterior Superior
Pancreaticoduodenal a.
Gastroduodenal a.

Perforated Peptic Ulcer

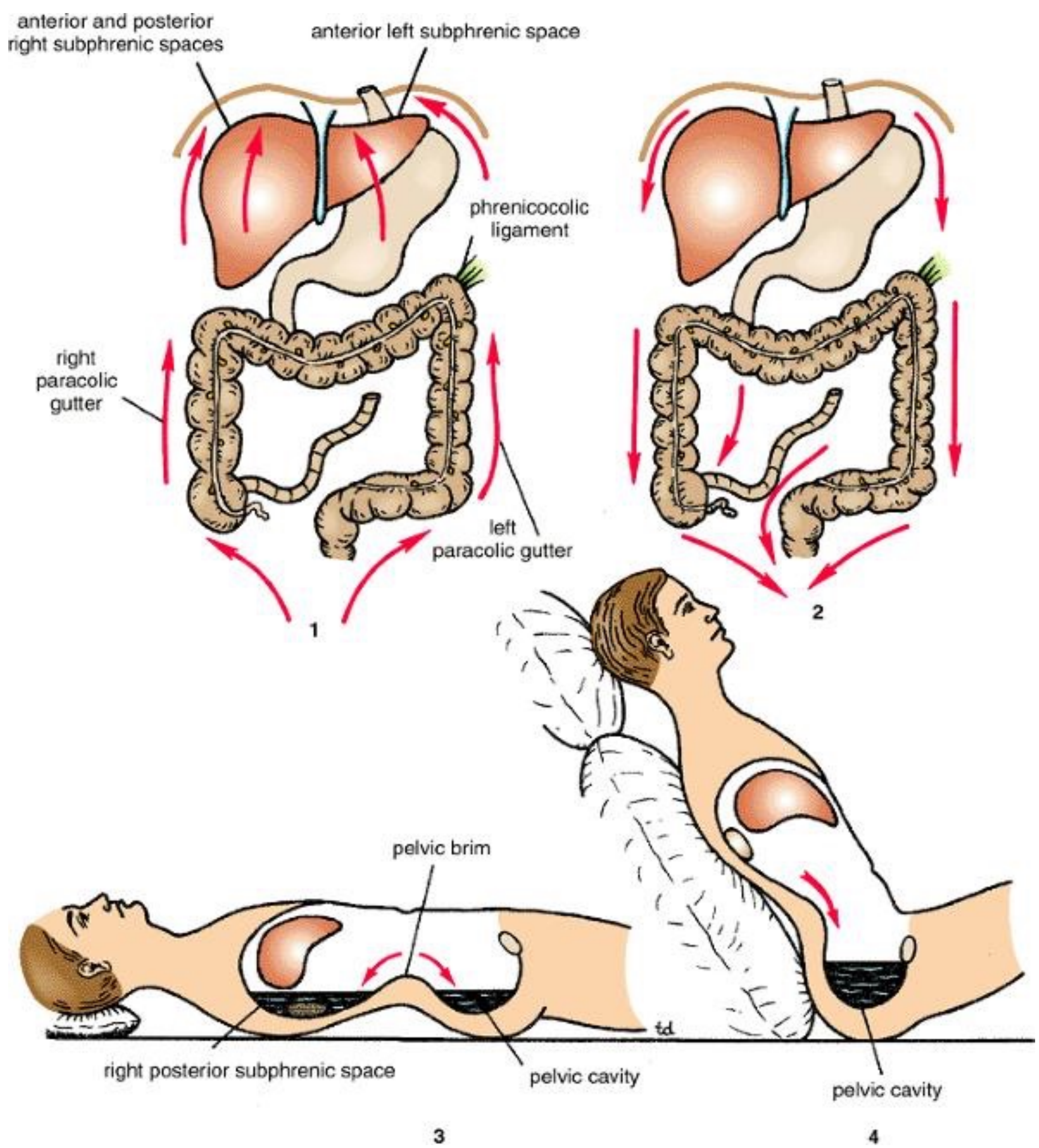




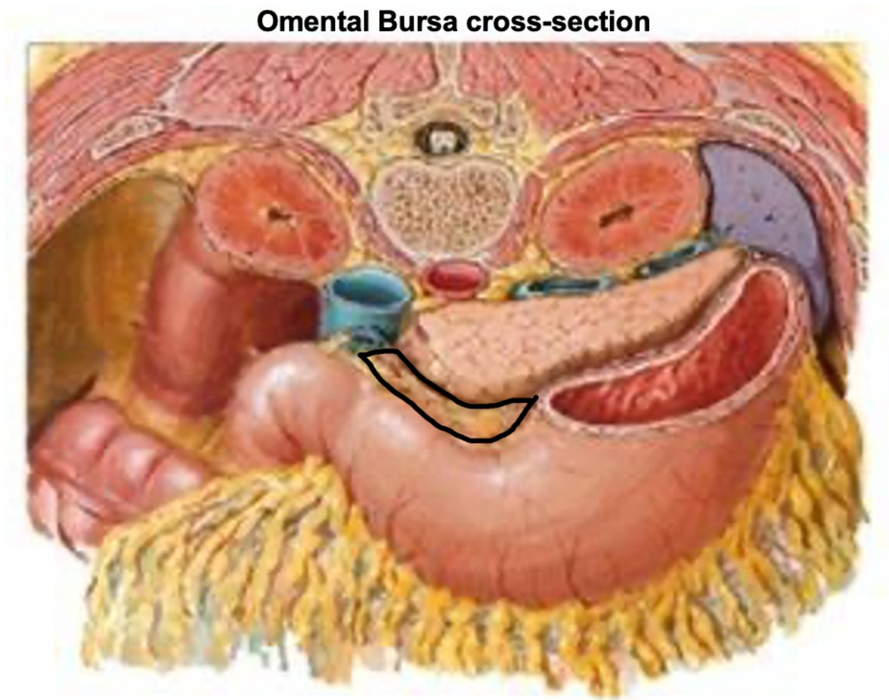
Clinical Presentation

- **Sudden** onset, **severe pain** in the **abdomen**
- **Board** like **rigidity of** the **abdomen** = **peritoneal irritation**
- Abnormal finding of **free air under** the **diaphragm**
- **If ulcer** has **perforated into** a **surrounding vessels**, excessive **bleeding** may cause **hemodynamic compromise**

Direction of Peritoneal Fluid Flow



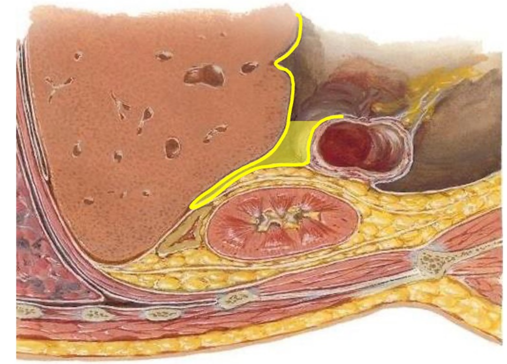
Perforation of the Stomach



Importance of Lesser sac

- **Perforation @ Posterior** wall of **Stomach (e.g., gastric ulcer)** can lead to **spilling** of **gastric contents**, a spilling event **can induce peritonitis** and **pancreatitis**

Perforation of the Stomach



Posterior Perforation

- Contents **spill into** the **Lesser sac (omental bursa)**
- If patient turns **Left Decubitus** = contents **remain in lesser sac**
- If patient turns **Right Decubitus** = flow from **lesser sac to greater sac**
- If patient **Supine** = **Hepatorenal recess + Subphrenic spaces**
- If patient **Erect** = flow **Paracolic gutters to Pelvis (rectouterine pouch, rectovesical pouch)**

Anterior Perforation

- Contents spill into the **Greater sac**

Peptic Ulcers

Duodenal ulcers

- **Most common** occurrence is **first part of** the **Duodenum (Duodenal bulb)**

- Flow of **acid through** the **pyloric valve** is **directed toward** the **posterior wall of** the **bulb**

- **Deep ulceration** can **invade** the **Gastroduodenal** artery **OR its branches** (**Posterior Superior Pancreaticoduodenal** artery)

- **ANTERIOR Duodenal** ulcers = **Peritoneal Cavity** = **Peritonitis**

Stomach

- The **secondary site of occurrence** is along the **Lesser Curvature of the stomach.**

- **Deep ulceration** can **invade** typically the **Left Gastric** artery, (**Splenic** artery **if posterior wall** – however is **uncommon**)

Case 2

A 65-year-old woman comes to the urgent care because of abdominal pain, nausea, loss of appetite and mild fever for the last 24 hours. The pain was initially dull and periumbilical but is now in the right lower quadrant. Blood pressure is 126/86mmHg, pulse is 68/min, respirations are 16/min and temperature is 36ºC (96.8ºF). Physical examination shows rebound tenderness in the right lower quadrant (RLQ) and a positive psoas sign. CT shows a metal-density foreign body in the right lower quadrant. Patient is referred for surgical treatment of suspected appendicitis.

1. Explain the reason for the change in the patient's pain complaint.

2. Explain the following signs and why they can be positive during appendicitis
   a. Psoas sign
   b. Rebound tenderness

3. Describe Mc Burney's point and explain its significance during an appendectomy.

Appendix

Dull Periumbilical Pain

- Embryologically, the **appendix** is **innervated by T10**, **as is** the **Periumbilical region**.

- Initial **referred pain in** the **Periumbilical region indicates** potential **appendicitis** is due to **Visceral Afferent** fibers **carrying pain from** the **appendix** relays **to DRG @ T10 level**

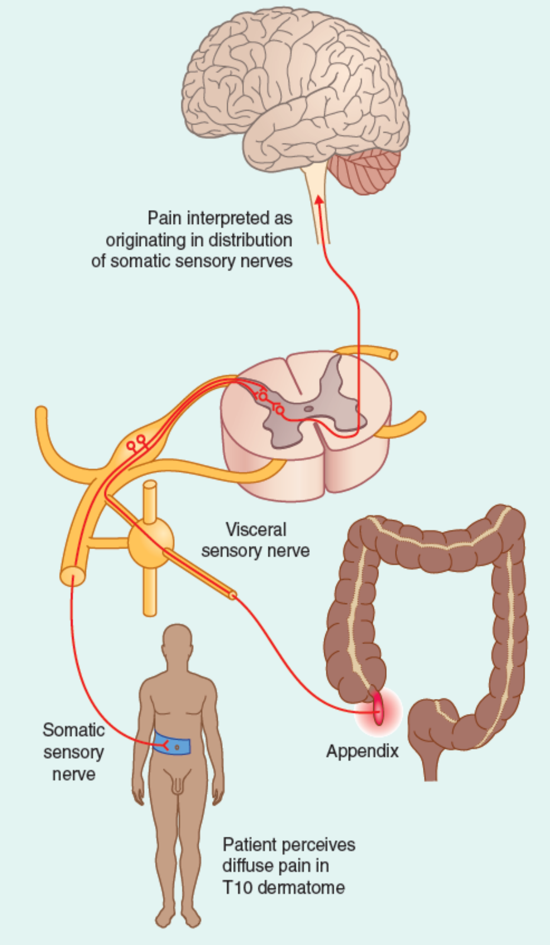


Fig. 1.50 Mechanism for referred pain from an inflamed appendix to the T10 dermatome.

Sharp RLQ Pain

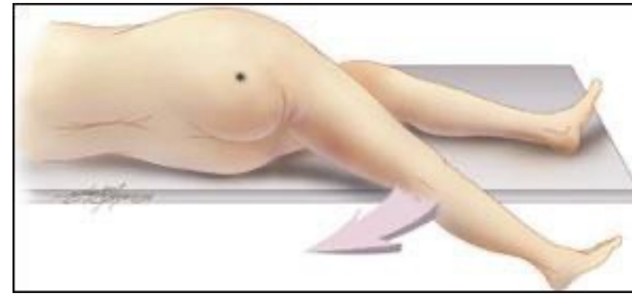
- **When** the **inflamed** the **appendix** comes **in contact with** the **body wall**, **pain** is usually **felt** in the **RLQ**. This pain will be **sharp because** of **Somatic Afferent** fibers **carrying pain** signals **from** the **parietal peritoneum** (**body wall**) **to the DRG at** the **associated dermatome** level.

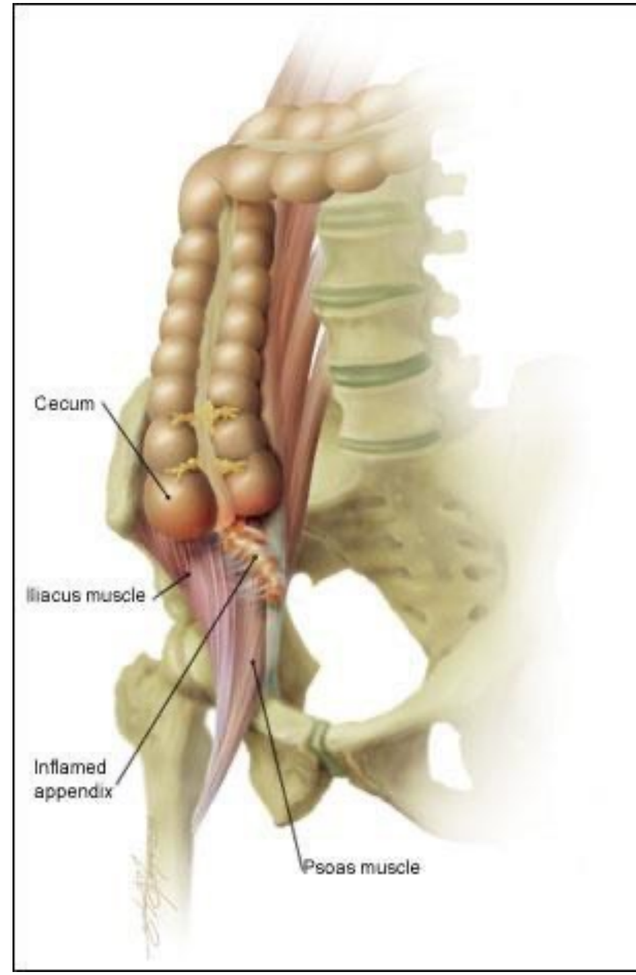
Psoas Sign

- **Psoas sign** is **used in** cases of **suspected appendicitis.**

- **Psoas sign** is **elicited by passively extending** the **thigh** of a patient lying on their left side **with knees extended or asking** the **patient to** actively **flex** their **thigh** at the hip **against resistance**. **If abdominal pain** occurs, this is a **positive psoas sign** or test.

- **Pain** occurs **because** the **psoas** muscle is being stretched (hyperextension at hip) **or** is contracting (flexion at hip), **causing friction against nearby inflamed tissues**.



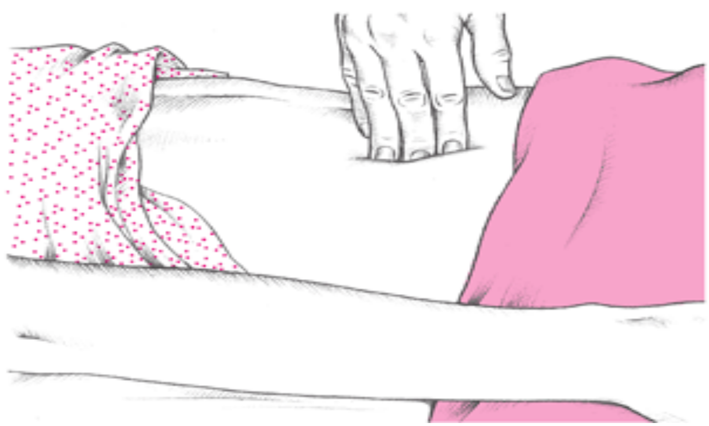


Rebound Tenderness

- **To confirm** the diagnosis of **appendicitis**, the **press down slowly on** the **area of tenderness**, and then **quickly withdraw hand**, rapidly releasing the pressure applied to the area (e.g., **RLQ**). **Patient** should be **asked if pain worse** on **pressing OR withdrawal** of the hand.

- **Because** of the **peritoneal irritation**, the **patient** may **experience sudden**, **sharp pain on** the **right side**, **in** the **region of infected appendix**.

Eliciting rebound tenderness



To elicit rebound tenderness, help the patient into a supine position and push your fingers deeply and steadily into his abdomen (as shown). Then quickly release the pressure. Pain that results from the rebound of palpated tissue—rebound tenderness—indicates peritoneal inflammation or peritonitis.

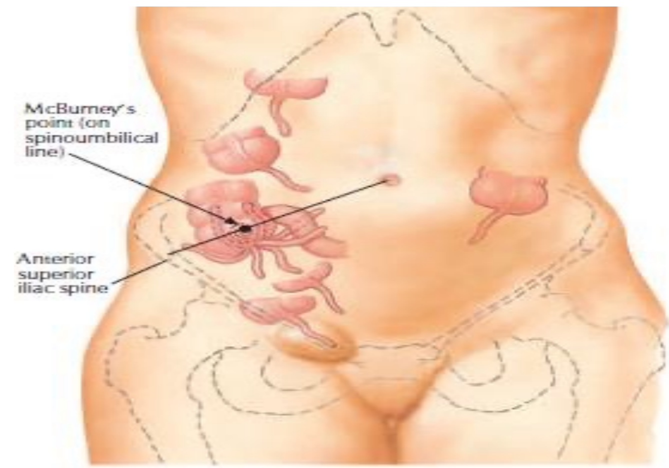
Appendix + McBurney's Point

- The **appendix** lies **in** the **right iliac fossa**. In relation to the anterior abdominal wall, its **base** is **situated 1/3 the way up McBurney's point** (**line joining** right anterior superior iliac spine **ASIS to Umbilicus**)

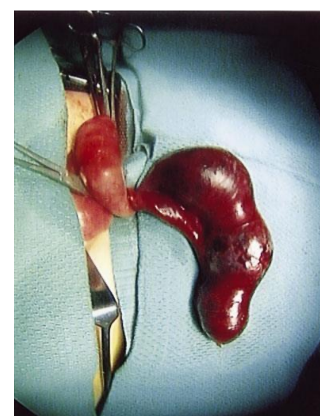
- Its **base** is **attached to** the **posteromedial surface of cecum** and **below ileocecal junction.**

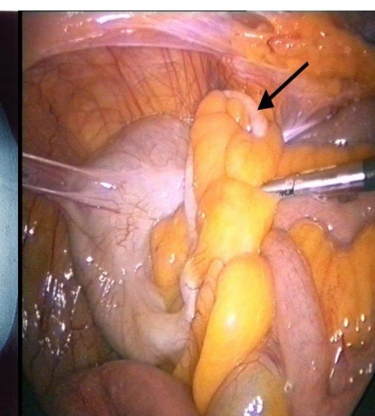
- **Completely surrounded by peritoneum (mesoappendix)**

- Blood **supply** is derived **from** the **appendicular artery from** the **ileocolic artery**



Variations in position of appendix



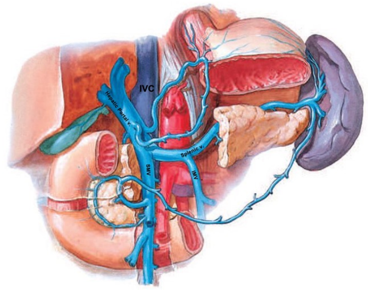


Case 3

A 45-year-old man comes to the emergency department because of two episodes of bloody vomiting over the past week. Patient also complains of dizziness and melena (dark, tarry stool). Patient has a history of chronic alcohol use. His blood pressure is 138/88mmHg, pulse is 104/min, respirations are 16/min and temperature is 38.9ºC (102ºF). Lab tests show low hemoglobin (anemia). Endoscopy confirms esophageal varices. CT scan shows enlargement of the liver and spleen, as well as dilated and tortuous splenic and portal veins. After additional testing, a diagnosis of portal hypertension is confirmed.

1. Define hematemesis and hematochezia.
2. Describe the portal circulation and the areas of anastomosis with the systemic circulation.
3. Describe portal hypertension and its relationship to esophageal and rectal varices, and caput medusa.
4. Describe how portal hypertension can be reduced.

Hepatic Portal Vein – Portal Circulation



- **Collects venous** blood **with products of - Digestion** from abdominal GI, **Gallbladder**, **Spleen** & **Pancreas**

- **Behind** the **neck of Pancreas** formed by **by joining** the **Splenic** vein **& SMV**

- **Passes through** the **Lesser omentum** (**hepatoduodenal ligament**), **enters** the **Liver** through ***Porta Hepatis***, **divides** into **Right & Left** and **then Segmental** branches, **ends in Venous Sinusoids of Liver**

Portal Venous Drainage

Hepatic Portal vein

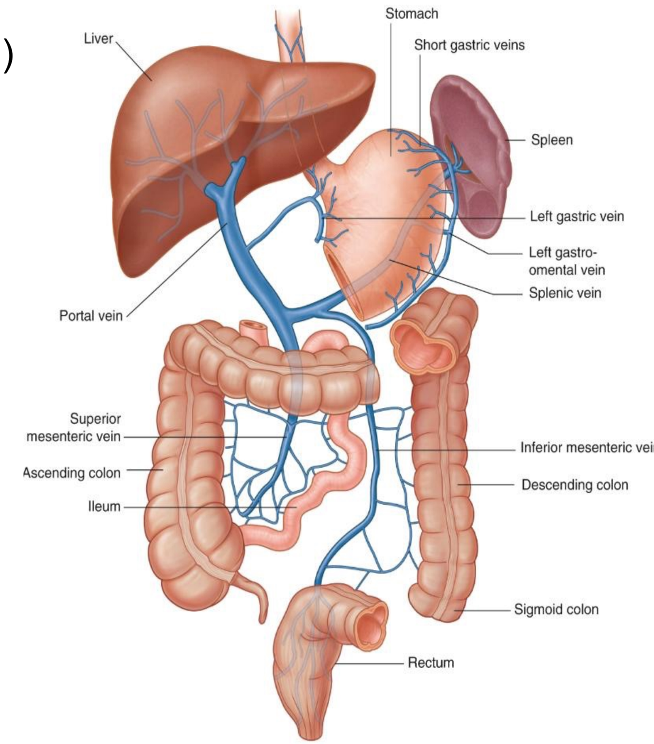
- **Left** & **Right Gastric** veins
- **Cystic** vein
- **Paraumbilical** veins (assoc.w/ obliterated umbilical v.)
- **Post. Superior Pancreaticoduodenal** (usually)

Splenic vein

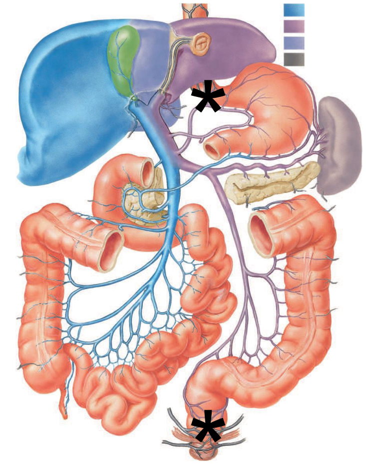
- **Short Gastric** veins
- **Left gastro-omental**/epiploic vein
- **Pancreatic** veins (great and dorsal)
- **Inferior Mesenteric Vein**

Superior Mesenteric Vein

- **Right gasro-omental/**epiploic vein
- **Ant. Superior pancreaticoduodenal** (usually)
- **Ant. + Post. Inferior pancreaticoduodenal**



Portocaval Anastomoses



Portal vein branches **communicate with** the **Caval** system

Lower end of Esophagus

Esophageal branches **of Left Gastric vein** anastomoses **WITH**
Esophageal branches **of Azygos & Hemiazygos veins**

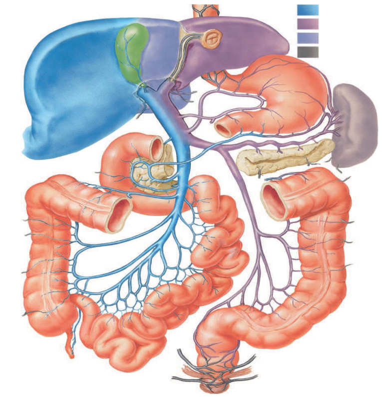
- **Esophageal varices** = **Hematemesis**

Lower end of Anal Canal

Superior Rectal vein (drains **into Inferior Mesenteric** vein) **WITH**
Middle & **Inferior Rectal** veins (drain **into internal iliac** vein)

- **Varicosity** = **Hemorrhoids (piles) Hematochezia**

Portocaval Anastomoses



Umbilicus [Anterior Abdominal Wall]

Para-umbilical veins (**veins along** the **ligamentum teres**) anastamose **with Superficial** branches **of Superior + Inferior Epigastric veins** & **especially Superficial Epigastric veins**.

Dilated tortuous veins radiating **from** the **umbilicus** are **called Caput Medusa**

Retro-peritoneal organs

Veins of **Colon**, **Duodenum**, **Pancreas**, **Spleen** etc. **WITH Renal**, **Lumbar**, **Azygos** veins etc.

- **Dilated veins** = **around** the **posterior abdominal wall**

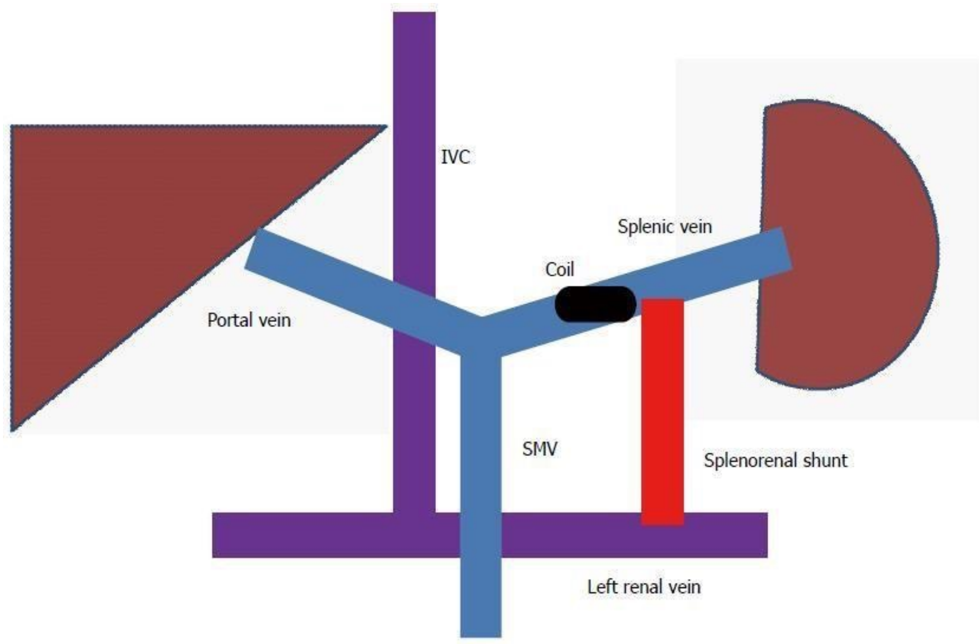
Bare area of the Liver

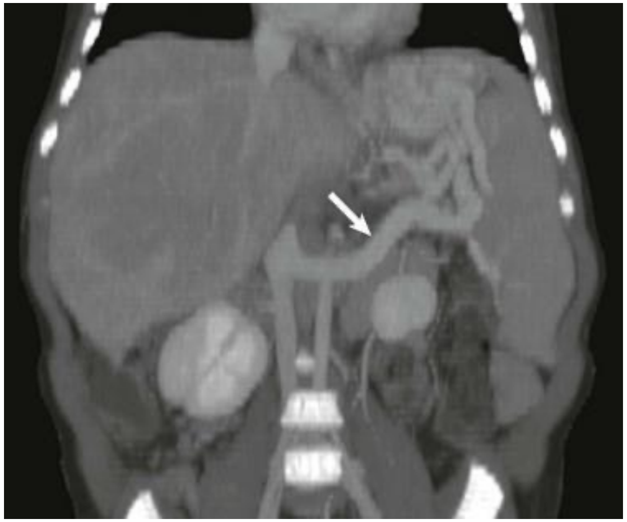
Venous sinusoids of **Liver WITH Diaphragmatic** veins (**intercostal** & **phrenic)**

Portal Hypertension

- Any pathology which **obstructs** the **hepatic portal** vein **increases pressure in** this **vein and** its branches (e.g., **cirrhosis**, vein **thrombosis**).

- A common **method for reducing portal hypertension** is to **divert blood from** the **portal** venous system **to** the **systemic Caval** venous system **by creating a shunt between** the **Portal vein** & **IVC**





e.g., of a portocaval anastomosis of portocaval shunt created by joining the splenic and left renal veins

Case 4

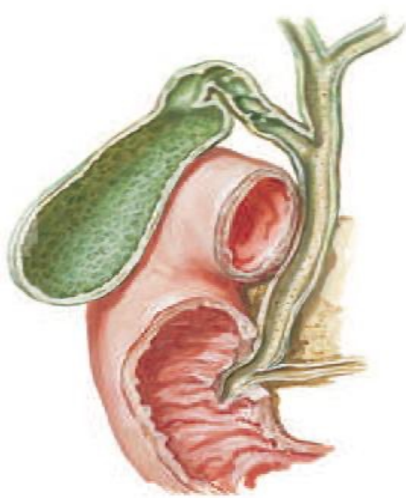
A 73-year-old woman comes to the emergency department because of nausea and vomiting after meals for the past 2 days. Patient also complains of loss of appetite, bloating and mild epigastric discomfort and right shoulder pain. Her blood pressure is 124/78mmHg, pulse is 104/min, respirations are 16/min and temperature is 38ºC (100.4ºF). Physical exam shows yellowing of the eyes. Laboratory tests show elevated serum bilirubin. Due to symptoms and lack of findings on US, CT and inflamed gallbladder on MRCP, patient is scheduled for Endoscopic retrograde cholangiopancreatography (ERCP) to confirm and treat the suspected obstruction within the biliary tree.

1. Define and explain Endoscopic Retrograde CholangioPancreatography (ERCP)
   .
2. Explain why visceral pain from an inflamed gallbladder can be referred to the shoulder.

3. List the 3 common sites where gall stones can be impacted and explain the effects of each on bile flow.

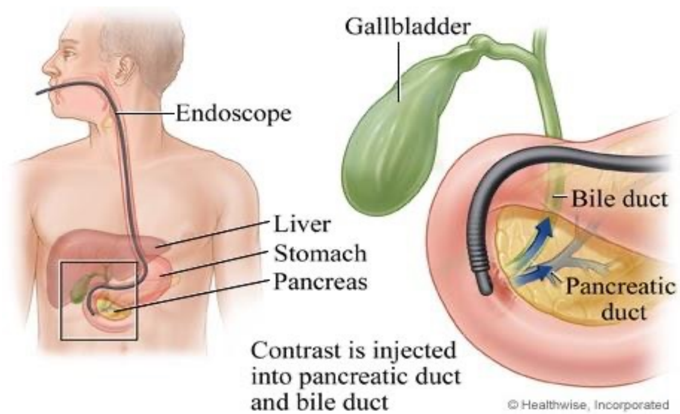
Biliary Apparatus

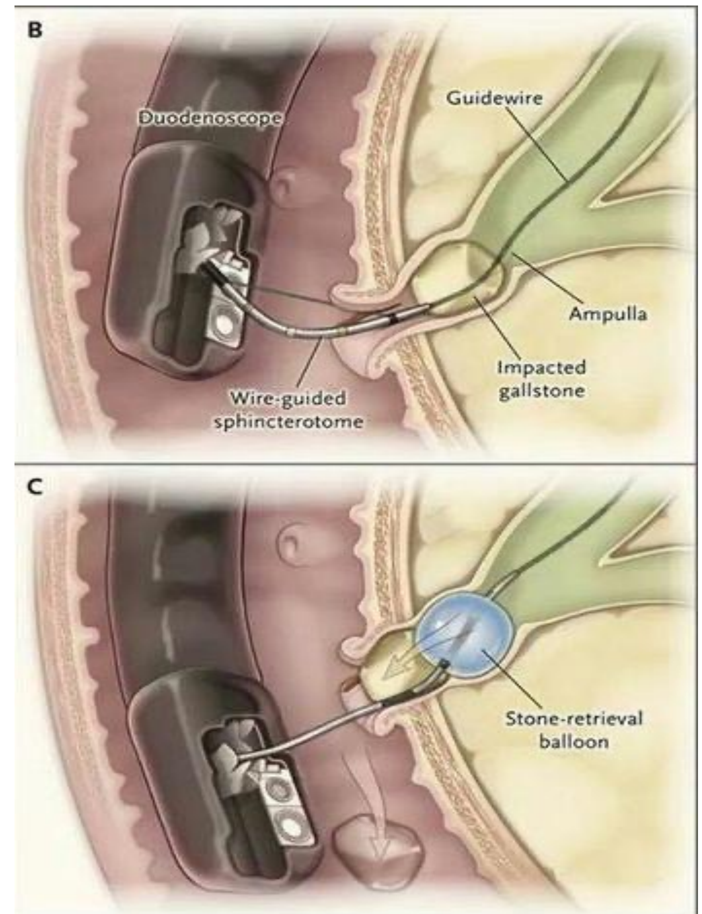
- **Right** & **Left Hepatic ducts** will **join** to **form** the **Common Hepatic** duct
- **Joined by** the **Cystic** duct **to form** the Common Bile duct
- The **Common Bile** duct **passes posterior to** the **1st part of Duodenum**
- **Joins** the **Pancreatic duct** to **open into** the **major Duodenal papilla**

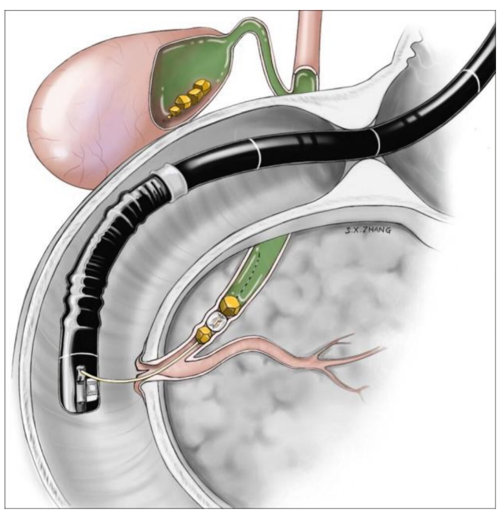


Endoscopic Retrograde Cholangio-Pancreatography (ERCP)

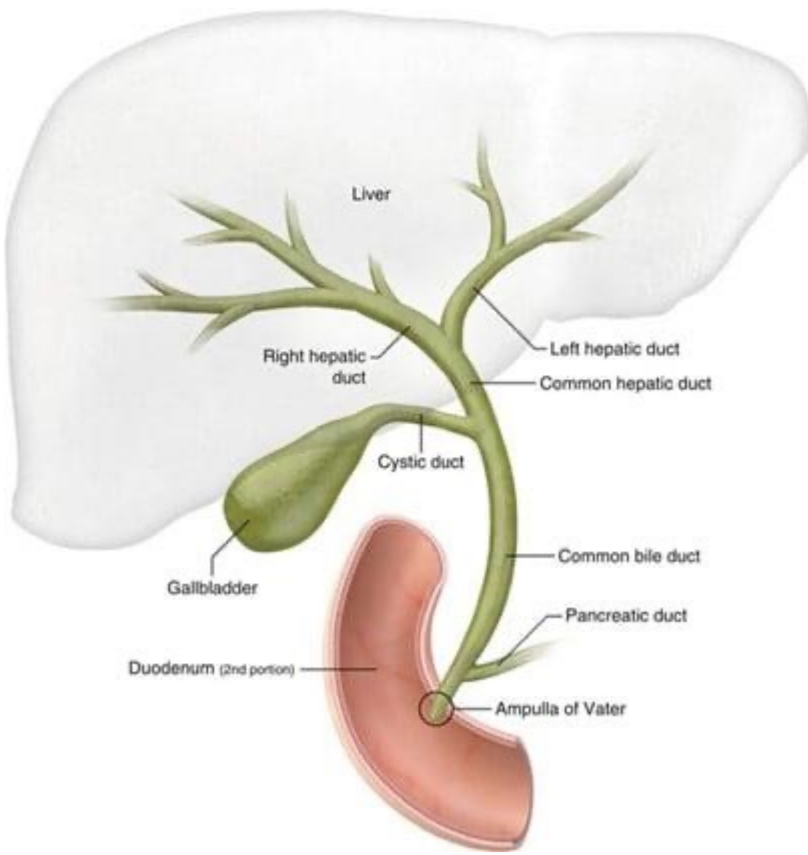
ERCP (endoscopic retrograde cholangiopancreatography) is a **diagnostic procedure** used to **evaluate** the **Biliary** & **Pancreatic ducts**. A **tube** is **passed via** the **esophagus**, **stomach** and **2nd part** of the **duodenum**. **Dye** is **injected into** the **major papilla** under **fluoroscopy**.







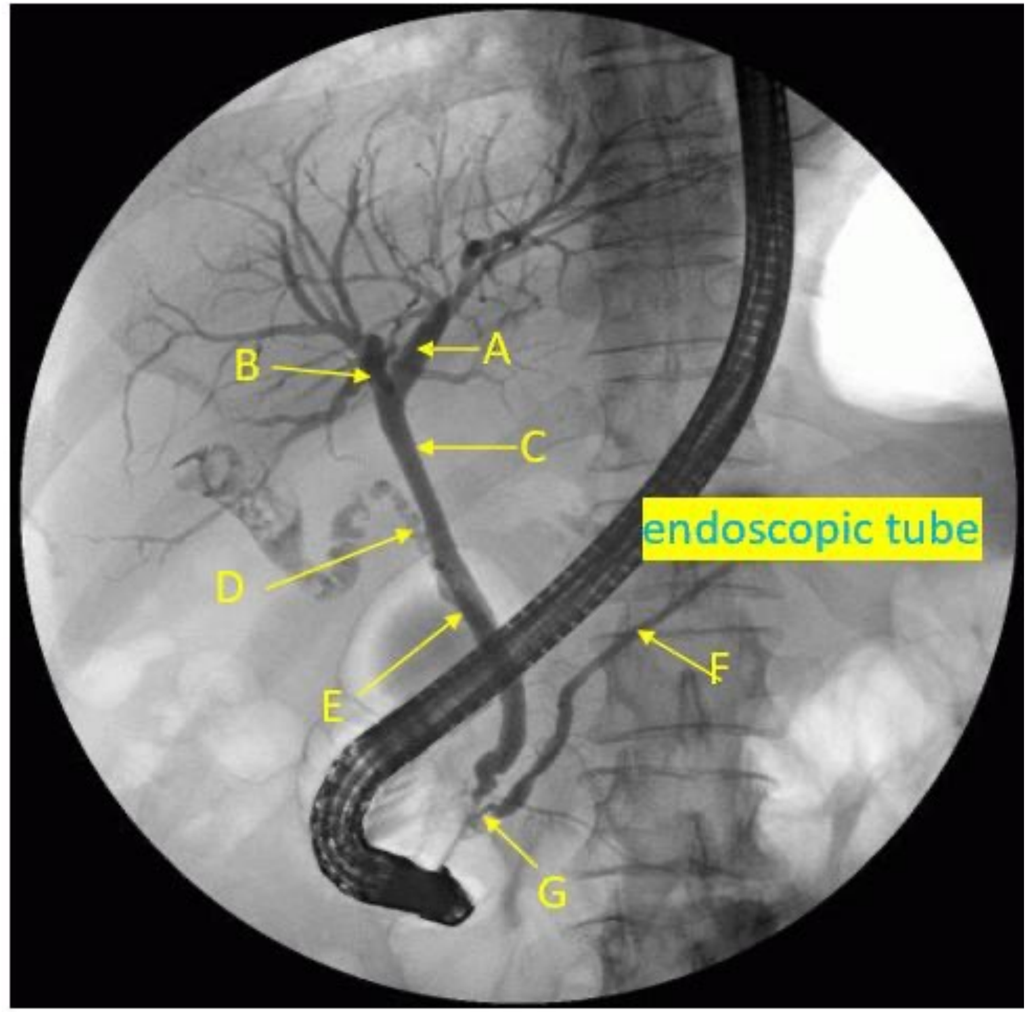
ERCP


Liver
Right hepatic duct
Left hepatic duct
Common hepatic duct
Cystic duct
Gallbladder
Common bile duct
Pancreatic duct
Duodenum (2nd portion)
Ampulla of Vater

ERCP

Identify

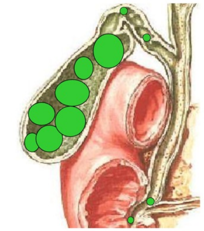
A. Left hepatic duct
B. Right hepatic duct
C. Common hepatic duct
D. Cystic duct
E. Common bile duct
F. Major pancreatic duct
G. Ampulla of Vater (hepatopancreatic ampulla)



Other structures:

- Liver
- Gall bladder
- 2nd part of duodenum

Gallstones



- **Gallstones** may be **asymptomatic** (common in **fat**, **fertile**, **female 40+)** or can **produce colic OR acute cholecystitis**

- **Biliary Colic** = usually **caused by spasm of** the **smooth** muscle of the **gallbladder** in an **attempt to expel** the **gallstones**

- **Acute Cholecystitis** = **pain** in the **RUQ =** may **cause Subdiaphragmatic Parietal Peritoneum irritation**, which is **supplied by** the **Phrenic** nerve (**referred pain** in **RIGHT shoulder**)

- **Obstruction of Biliary tree** obstructed **by Gallstone OR compression by Pancreatic growth** may lead to obstructive **jaundice**. Impaction of **stone in** the **Ampulla** can **cause** passage of infected **bile into** the **Pancreatic duct** leading to **Pancreatitis**